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1.

Knowledge representation is the process of storing information in a form that an AI system can understand and use to make decisions. A good example is a chatbot used for IT support. The chatbot can store knowledge about common computer problems, their causes, and possible solutions. For example, the system can represent the relationship between a problem and its solution using rules such as: “If a computer has no internet connection, check the network cable or Wi-Fi connection.” The chatbot can also represent information about users, devices, software, and network services. When a user describes a problem, the chatbot compares the information provided with the knowledge stored in its system. It can then use the relevant rules to determine an appropriate response. Knowledge representation allows the chatbot to organize information and reason with it instead of simply storing random data. This makes the chatbot more useful because it can use existing knowledge to provide consistent answers and assist users with common IT problems.

2.

First-Order Logic (FOL) is a formal method used in AI to represent knowledge about objects, their properties, and relationships. Unlike simple propositional logic, FOL can describe individual objects and how they are related to one another. It uses concepts such as variables, predicates, constants, and quantifiers to represent information. For example, the statement “All computers need an operating system” can be represented using a rule that applies to every computer. AI systems can use First-Order Logic to store facts and rules and reason about them to reach conclusions. It is useful in expert systems, knowledge-based systems, natural language processing, and automated reasoning. FOL helps AI represent more complex information in a structured and logical way.

3.

For the tool practice, I created a basic AND gate truth table in Logisim. An AND gate has two inputs, A and B, and produces an output of 1 only when both inputs are 1. I tested all four possible combinations of the two inputs. When both inputs were 0, the output was 0. When A was 0 and B was 1, the output remained 0. When A was 1 and B was 0, the output was also 0. Finally, when both A and B were 1, the output became 1. This activity helped me understand how logical operations work in digital systems. It also showed me

how truth tables can represent knowledge about possible inputs and their expected outputs. Logisim made the relationship between the inputs and output easy to observe.

4.

A semantic network is a diagram used to represent knowledge by showing concepts and the relationships between them. For this diagram, the main concept can be an IT Support Chatbot. The chatbot can be connected to related concepts such as User, Computer, Problem, Software, and Solution. For example, a User can have a Computer, the Computer can have a Problem, and the Problem can have a Solution. The relationships can be represented using arrows between the different concepts. For example, “User → uses → Computer” and “Computer → has → Problem.” This creates a simple visual representation of how information is connected. The semantic network helps an AI system organize knowledge and understand relationships between different objects. It also makes complex information easier for people to understand because the relationships can be viewed directly in the diagram.

Example Truth Table — AND Gate

A	B	Output (A AND B)
0	0	0
0	1	0
1	0	0
1	1	1